

The Data Layer

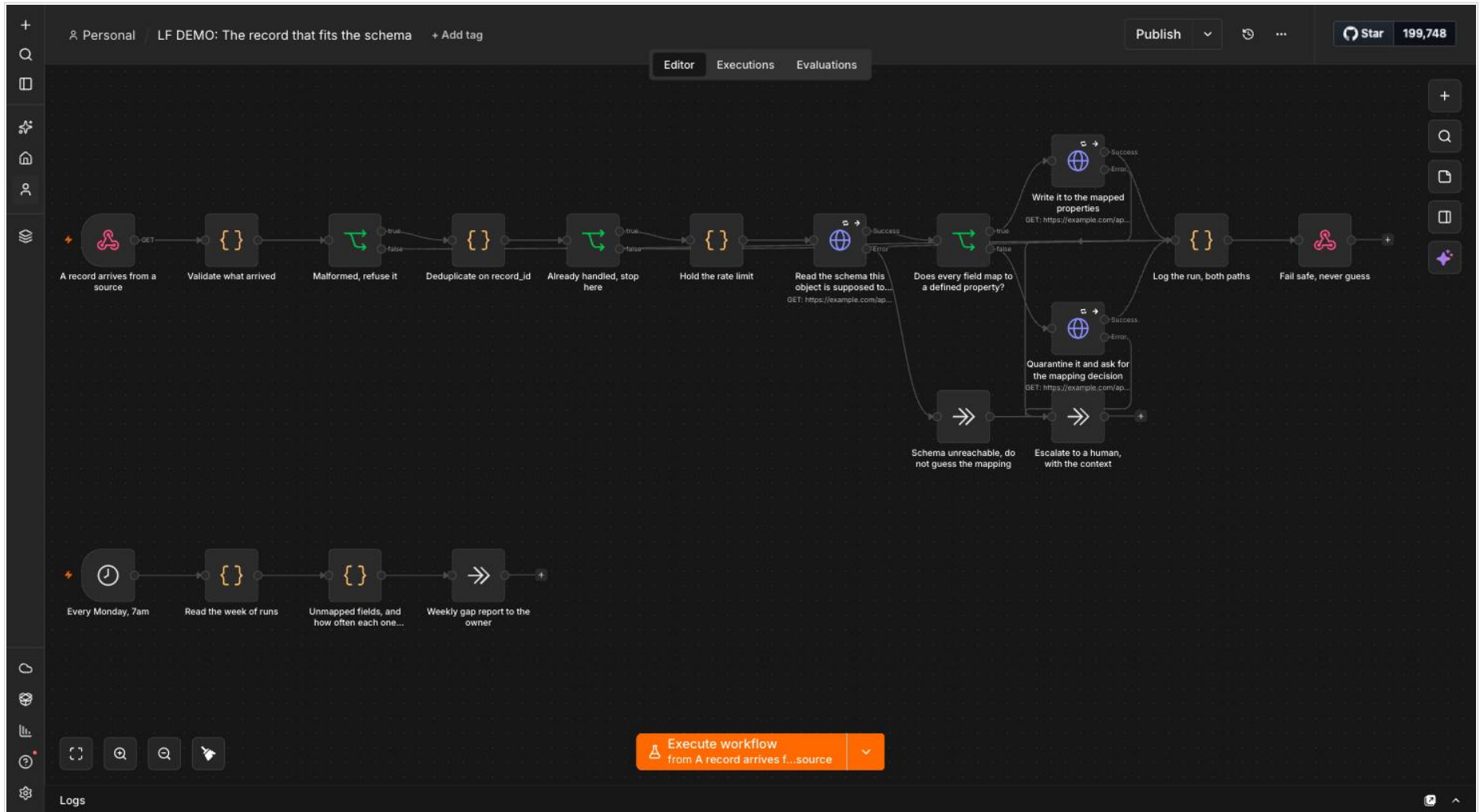
data-model-architecture · built in hubspot

Most CRM problems are data-model problems wearing a workflow costume.

What it connects to: The schema has to hold for every system that writes into it. A property that only makes sense in one tool is the one that breaks the reporting later.

1 of 4 SYSTEM

Here's the system that enforces it — a record only gets written once every field maps to a property that exists.



2 of 4 THINKING

This is the gate that keeps the model clean: every field has to map to a property that exists, or the record is quarantined rather than guessed at. Guessing is how you end up with three fields meaning the same thing.

The screenshot shows a dark-themed user interface for a data processing tool. The main window is titled "Does every field map to a defined property?". It is divided into two main sections: "INPUT" on the left and "Parameters" / "Settings" on the right.

INPUT Section:

- At the top, it says "INPUT".
- Below that, there is a large area with the text "No input data" and a button "Execute previous nodes" with the subtitle "to view input data".
- On the far left, there is a vertical sidebar with various icons, including a globe icon.

Parameters / Settings Section:

- At the top right of this section is an orange button labeled "Execute step".
- Below the button is a section titled "Conditions".
- Inside the "Conditions" section, there is a text area containing the following code:

```
{{ $json.fullyMapped === true && $json.objectExists === true }}
```
- To the right of the code area is a dropdown menu with the text "is true".
- Below the code area is a button labeled "+ Add condition".
- Below the "+ Add condition" button is a toggle switch labeled "Convert types where required", which is currently turned off.
- Below the toggle switch is a section titled "Options" with a plus sign icon to its right.
- Below the "Options" section is a button labeled "+ Add option".

3 of 4 THEIR-WORLD

Here's the object model on a real portal — objects, properties, and how they associate.

The screenshot displays the 'Data Model' section of a portal, specifically the 'Manage' tab. The interface is organized into several key areas:

- Top Bar:** Includes a 'Find or Ask' search bar, navigation icons, and links for 'Upgrade', 'Assistant', and 'Launch Forte'.
- Left Sidebar:** Labeled 'Data Model', it contains tabs for 'Intro', 'Manage' (active), 'Health', and 'Analysis'. Below these is a list of object categories: 'All objects', 'Calls', 'Carts', 'Communications', 'Companies', 'Contacts', 'Contracts', 'Credit memos', 'Deals', 'Emails', 'Invoices', and 'Line items'.
- Search and Filter Bar:** Features a 'Search for an object' input, a 'Viewing all objects' dropdown, and a '(2) Object status' dropdown. Action buttons for '+ Create object' and 'More' are also present.
- Object Grid:** The main area displays a grid of object cards. Each card represents an object type (e.g., 'Appointments', 'Carts', 'Companies', 'Contracts', 'Courses', 'Listings', 'Marketing events', 'Orders', 'Projects', 'Services', 'Tickets', 'Users') and provides details such as the number of properties and associated objects. For example, 'Appointments' has 47 properties and 14 associated objects. Each card includes an 'Activate object' button and a 'View all' link.
- Right Panel:** Contains a 'Self-paced tours' dropdown, an 'Import' button, and a 'More' dropdown. A small 'Activities' section is visible at the bottom right, showing a list of activities like 'Calls', 'Communications', and 'Emails'.

And the property architecture underneath it, with fill rates so you can see what's actually used.

Find or Ask

Upgrade

Assistant

Launch Forte

Dashboard

Search Settings

Notifications

Account Management

Account Defaults

Account & Billing

Audit Log

Users & Teams

Product Updates

Integrations

Marketplace Downloads

Tracking & Analytics

Privacy & Consent

Security

AI

Payments Account

Data Management

Properties

Properties

Data Agent

Data Quality

Export properties

Properties are used to collect and store information about your records in HubSpot. For example, a contact might have properties like First Name or Lead Status.

Select an object:

Contact properties

Go to Contacts settings

Properties (211)

Conditional logic

Groups

Archived (0)

All groups

All field types

All users

All properties

Search properties

Data quality monitoring is off

Create property

	Name ↑	Group ↓	Created by ↓	Used In ↓	Fill Rate
☐	Annual Revenue Single-line text	Contact information	HubSpot	0	0%
☐	Average Pageviews Number	Web analytics history	HubSpot	0	100%
☐	Buying Role Multiple checkboxes	Deal information	HubSpot	0	0%
☐	Campaign of last booking in meetings tool Single-line text	Contact activity	HubSpot	0	0%
☐	Chat Assistant IQL Date Date and time picker	Contact information	HubSpot	0	0%